10. Categorical data analysis: Two-way classifications

Example 10.1: Cinema goers

	Male	Female
Cinema often	22	21
Not often	20	12

- A two-way classification arises when each item may be classified according to two separate criteria. We get a two-way table or cross-tabulation.
- Data can be qualitative, ordered, quantitative.
- Analysis the same – interpretation may be different.
- A two-way table is often called a 'contingency table'

Notation (2×2 case)

Consider a classification with one **factor** having 2 **levels** (F_1 and F_2), and two different responses (R_1 and R_2):

	F_1	F_2	Total
R_1	n_{11}	n_{12}	$n_{1\bullet}$
R_2	n_{21}	n_{22}	$n_{2\bullet}$
Total	$n_{\bullet 1}$	$n_{\bullet 2}$	$n_{\bullet\bullet} = n$

This is the case in example 10.1.

Can also have a classification with 2 factors.

Denote by p_{ij} the probability that a sample unit belongs to row i and column j

Usual hypotheses

(a) Multinomial data (*table total n fixed*)

$$H_0\text{: variables/responses are independent}$$
$$\Pr(\text{row } i \text{ and column } j) = \Pr(\text{row } i) \times \Pr(\text{column } j)$$

That is,

$$H_0 \quad : \quad p_{ij} = p_{i\bullet} \times p_{\bullet j} \text{ for all } i, j$$
$$\text{versus } H_1 \quad : \quad p_{ij} \neq p_{i\bullet} \times p_{\bullet j} \text{ for some } i, j$$

(b) Condition on n and $n_{\bullet j}$ (*fixed column totals*)

$$N_{ij} \sim Bin\,(n_{\bullet j},\, p_{ij}) \quad j = 1, 2, \ldots, c; \text{ independent}$$

H_0: response is homogeneous ($p_{ij} = p_i$ for all levels j

i.e. null hypothesis is that response has the same distribution for all levels of the factor.

Analysis for both cases is the same – relies on the usual χ^2 statistic.

Test of no association (independence)

Test independence of response and factor (or two factors) using the Pearson χ^2 statistic:

$$X^2 = \sum_{i=1}^{r}\sum_{j=1}^{c} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

which under H_0 has a χ^2 distribution with $(r-1) \times (c-1)$ df.

Here: r is number of rows
c is number of columns
o_{ij} and e_{ij} are observed and expected frequencies.

In this context,

$$\boxed{e_{ij} = \frac{n_{i\bullet} \times n_{\bullet j}}{n}}$$

Example 10.2

A consumer association has done some research on customer views on the reliability of budget airlines. The results are tabulated below.

Airline	Good	Average	Poor
High life	50	40	30
Sky Flying	40	50	80
Up and Away	35	55	50

Carry out an appropriate hypothesis test to determine if there is an association between the airline and reliability.

H_0: There is no relationship between airline and reliability (i.e. the distributions are independent)

H_1: There is a relationship between airline and reliability

Example 10.2

First add marginal totals to the table.

Airline	Good	Average	Poor	Total
High life	50	40	30	120
Sky Flying	40	50	80	170
Up and Away	35	55	50	140
Total	125	145	160	430

Then calculate row and column probabilities and hence expected frequencies under H_0:

Airline	Good	Average	Poor	Total
High life	34.9	40.4	44.7	120
Sky Flying	49.3	57.2	63.5	170
Up and Away	40.8	47.4	51.8	140
Total	125	145	160	430

Example 10.2

$$\begin{aligned} \text{Degrees of freedom} &= (\text{rows} - 1) \times (\text{columns} - 1) \\ &= (3-1) \times (3-1) = 4 \end{aligned}$$

Value of test statistic is

$$X^2 = \frac{(50-34.9)^2}{34.9} + \frac{(40-40.4)^2}{40.4} + \cdots + \frac{(50-51.8)^2}{51.8} = 20.425$$

Under H_0 should be from χ^2 distribution with 4 df

p-value$= \Pr\left(\chi^2_4 \geq 20.425\right) = 0.0004$

Strong evidence against H_0. Seems reliability is not independent of airline.

In `R`, can use `chisq.test`

Yates' continuity correction

- Used in 2×2 contingency tables for tests of independence.
- We want to apply correction for approximating binomial probabilities with a continuous χ^2 distribution.
- Correction:

  Subtract 0.5 from $|o - e|$ before squaring it.

- This produces p-values that are closer to an 'exact' solution.
- But can be overly conservative.

Example 10.3

In a clinical trial, 50 patients are given a drug (treated group) and 50 are given a placebo (control group); the table gives the numbers suffering particular side-effects.

	Drug	Placebo	Total
Side-effects	15	4	19
No side-effects	35	46	81
Total	50	50	100

H_0: Suffering side effects is independent of choice of drug or placebo

H_1: Not independent

Example 10.3

Expected frequencies computed under H_0 are

	Drug	Placebo	Total
Side-effects	9.5	9.5	19
No side-effects	40.5	40.5	81
Total	50	50	100

Under H_0, test statistic is from χ^2 distribution with 1 df

Without Yates' correction, test statistic is $X^2 = 7.8622$, and so p-value $= 0.005$

With Yates' correction, test statistic is $X^2 = 6.4977$, and so p-value $= 0.0108$

In this case, the evidence against H_0 is less strong with correction than one would think without correction.

Fisher's exact test

- With small data sets the χ^2 approximating distribution is not valid.
- For 2×2 tables we can use **Fisher's test** to calculate exact p-values.
- It relies on the *hypergeometric distribution* for the counts in the table when the row and column margins are considered to be fixed.

Fisher's exact test

Consider the following general case:

	F_1	F_2	Total
R_1	a	b	$a+b$
R_2	c	d	$c+d$
Total	$a+c$	$b+d$	$n=a+b+c+d$

Suppose the smallest observed count is b, say. Then exact probability for observed value b is

$$\Pr(B=b) \quad = \quad \frac{\binom{a+b}{b}\binom{c+d}{d}}{\binom{n}{b+d}} \quad = \quad \frac{(a+b)!\,(c+d)!\,(a+c)!\,(b+d)!}{a!\;b!\;c!\;d!\;n!}$$

Fisher's exact test

The p-value is then given by the probability to observe a value at least as extreme as b

$$\begin{aligned} \text{p-value} &= \Pr(B \leq b) \\ &= \Pr(B = b) + \Pr(B = b-1) + \cdots + \Pr(B = 0) \end{aligned}$$

We can get a 2-sided p-value as

$$\text{p-value} = \Pr(B \leq b) + P(B \geq a)$$

Example 10.3 (continued)

Clinical trial data.

	Drug	Placebo	Total
Side-effects	15	4	19
No side-effects	35	46	81
Total	50	50	100

$$\Pr(B = 4) = \frac{\binom{19}{4}\binom{81}{46}}{\binom{100}{50}} = \frac{19!\ 81!\ 50!\ 50!}{15!\ 4!\ 35!\ 46!\ 100!} = 0.0039$$

$$\text{p-value} = \Pr(B \leq 4) + \Pr(B \geq 15) = 0.0047 + 0.0047 = 0.0094$$

We have evidence against independence.

Observe that the exact p-value here is close to the p-value of the χ^2 test with Yates' correction.

In `R`, can use `fisher.test`

Example 10.1 (continued)

Cinema goers.

	Male	Female	Total
Cinema often	22	21	43
Not often	20	12	32
Total	42	33	75

$$\begin{aligned} \Pr(\text{Often} \mid \text{Male}) &= 22/42 = 0.524 \\ \Pr(\text{Often} \mid \text{Female}) &= 21/33 = 0.636 \end{aligned}$$

Significant difference?

Is there an association between gender and cinema attendance?

H_0: No association between gender and cinema attendance.

Example 10.1

Carry out χ^2 test in R.

```
> matcinema=matrix(c(22,20,21,12),2,2)
> chisq.test(matcinema)

        Pearson's Chi-squared test with Yates' continuity correction

data:  matcinema
X-squared = 0.5522, df = 1, p-value = 0.4574
```

No evidence of association between gender and cinema attendance.

```
> chisq.test(matcinema)$expected
      [,1]  [,2]
[1,] 24.08 18.92
[2,] 17.92 14.08
```

Example 10.4

Consider now a larger sample with same proportions.

	Male	Female	Total
Cinema often	110	105	215
Not often	100	60	160
Total	210	165	375

$$\begin{aligned} \Pr(\text{Often} \mid \text{Male}) &= 110/210 = 0.524 \\ \Pr(\text{Often} \mid \text{Female}) &= 105/165 = 0.636 \end{aligned}$$

Significant difference?

Perform χ^2 test in R

Example 10.4

```
> matcinema2=matrix(c(110,100,105,60),2,2)
> chisq.test(matcinema2)

        Pearson's Chi-squared test with Yates' continuity correction

data:  matcinema2
X-squared = 4.3361, df = 1, p-value = 0.03731
```

Evidence (at 5% level) to reject H_0

Evidence that there is an association between gender and cinema attendance.

```
> chisq.test(matcinema2)$expected

      [,1] [,2]
[1,] 120.4 94.6
[2,]  89.6 70.4
```